

Lab Assignment #6

Part 1.

Unification (8 points). For each of the following pairs, state whether or not they will unify. In each case, state what the most general unifier is, or show that unification is impossible. If they unify, show the unification of the two terms. In this problem, U, X, Y, Z are variables and a, b, c, d are constant symbols. (This is a paper-and-pencil problem, and no programming is required.)

- I. $p(b, b, a)$ and $p(X, Y, Z)$.
- II. $p(f(a, b), c, d)$ and $p(f(X, Y), X, Z)$.
- III. $p(f(a, b), a, d)$ and $p(f(X, Y), X, Z)$.
- IV. $p(a, Z, Z)$ and $p(Y, Y, b)$.
- V. $q(X, f(X, Y), Y)$ and $q(U, f(Z, c), Z)$.
- VI. $q(f(X, Y), Y, X)$ and $q(Z, U, U)$.
- VII. $q(X, Z, Z)$ and $q(U, U, Y)$.
- VIII. $q(X, f(Y, g(X, a)))$ and $q(h(U, a), Z)$.

Part 2

Create a Prolog database for an airline company. The database consists of ground¹ unit clauses made from the following predicates:

```
plane(plane#, model, age)
flight(flight#, from, to, plane#)
passenger(sin, name, city, age)
booked(sin, flight#, date, price)
```

These predicates are interpreted as follows. Each plane has a unique plane number (plane#), a model, and an age. Each flight has a unique flight number (flight#) and flies from one city to another. A flight may fly between several cities (eg, from Toronto to Paris, and then from Paris to Frankfurt.) Each flight is assigned a unique plane, identified by its plane number. Each passenger has a name, a city, an age, and a unique social insurance number (sin). Many passengers may have the same name and city. Passengers are booked on a flight on a given date for a given price. A passenger may be booked on many flights, even on the same day.

Here are typical ground unit clauses for each of the four predicates:

```
plane(p123,boeing747,8)
flight(f407,toronto,paris,p123)
passenger(s456_234_987,anna,toronto,26)
```

¹ A ground predicate is one that contains no variables

booked(s456_234_987,f407,jan17,500)

- The first means that plane p123 is a Boeing 747 and is 8 years old.
- The second means that flight f407 goes from Toronto to Paris and is assigned plane p123.
- The third means that Anna lives in Toronto, is 26 years old, and her social insurance number is s456 234 987.
- The fourth means that person s456 234 987 is booked on flight f407 on January 17, and she paid \$500 for the booking.

Create a Prolog database of at least 20 ground unit clauses using the four predicates above (about 5 per predicate). Choose the data so that the following queries return interesting non-trivial answers (e.g., not always an empty answer, not always the same answer, not always a single answer).

Write Prolog queries for the following :

- (a) The model and age of plane p123.
- (b) The plane number and age of every boeing 747.
- (c) The flight number of every plane going from New York to Paris.
- (d) The flight number and price paid by every passenger going from Los Angeles to Paris on Jan 17.
- (e) The name and SIN of every passenger booked on flight f237 on Feb 23.

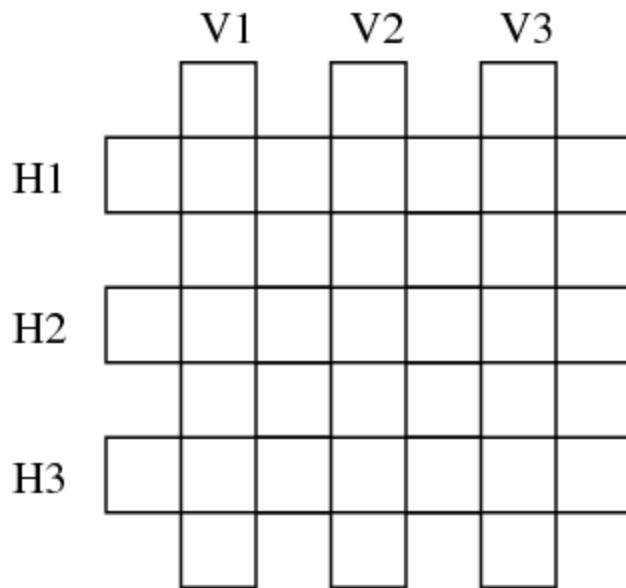
Submit the pen and paper exercise as a pdf and the Prolog program (including queries) as a .pl file.

Part 3

Here are six Italian words:

astante , astoria , baratto , cobalto , pistola , statale .

They are to be arranged, crossword puzzle fashion, in the following grid:



The following knowledge base represents a lexicon containing these words:

```
word(astante, a,s,t,a,n,t,e).
word(astoria, a,s,t,o,r,i,a).
word(baratto, b,a,r,a,t,t,o).
word(cobalto, c,o,b,a,l,t,o).
word(pistola, p,i,s,t,o,l,a).
word(statale, s,t,a,t,a,l,e).
```

Write a predicate `crossword/6` that tells us how to fill in the grid. The first three arguments should be the vertical words from left to right, and the last three arguments the horizontal words from top to bottom.

Submit the pen and paper exercise as a pdf and the Prolog programs (including queries) as .pl file files.